

Angle Modulation

Prepared by

Rifat Rahman

Assistant Professor

Institute of Appropriate Technology, BUET

<https://rrahman007.buet.ac.bd>

and

Prof. Hitesh Dholakiya

Spectral Components of Angle Modulation

□ Equation of Frequency Modulation {If the carrier signal is $V_c \sin(\omega_c t)$ and the Message Signal is $V_m \cos(\omega_m t)$ } is given by:

$$Y_{FM}(t) = V_c \sin(\omega_c t + m_f \sin(\omega_m t)) \text{ [Here, Modulation Index, } m_f = \frac{\Delta f}{f_m}]$$

$$Y_{FM}(t) = V_c \left[J_0(m_f) \sin(\omega_c t) + \sum_{n=1}^{\infty} J_n(m_f) (\sin((\omega_c + n\omega_m)t) + (-1)^n \sin((\omega_c - n\omega_m)t)) \right]$$

$$Y_{FM}(t) = V_c [J_0(m_f) \sin(\omega_c t) + J_1(m_f)(\sin((\omega_c + \omega_m)t) - \sin((\omega_c - \omega_m)t)) + J_2(m_f)(\sin((\omega_c + 2\omega_m)t) + \sin((\omega_c - 2\omega_m)t)) + J_3(m_f)(\sin((\omega_c + 3\omega_m)t) - \sin((\omega_c - 3\omega_m)t)) + J_4(m_f)(\sin((\omega_c + 4\omega_m)t) + \sin((\omega_c - 4\omega_m)t)) + \dots]$$

$$\sin(A + mB) = J_0(m) \sin A + \sum_{n=1}^{\infty} J_n(m) [\sin(A + nB) + (-1)^n \sin(A - nB)]$$

□ Here, $J_0(m_f), J_1(m_f), J_2(m_f), \dots$ are Bessel Function and their value depends on Modulation Index.

Bessel Function Table with respect to Modulation Index

Modulation index	Sideband																
	Carrier	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0.00	1.00																
0.25	0.98	0.12															
0.5	0.94	0.24	0.03														
1.0	0.77	0.44	0.11	0.02													
1.5	0.51	0.56	0.23	0.06	0.01												
2.0	0.22	0.58	0.35	0.13	0.03												
2.41	0	0.52	0.43	0.20	0.06	0.02											
2.5	-0.05	0.50	0.45	0.22	0.07	0.02	0.01										
3.0	-0.26	0.34	0.49	0.31	0.13	0.04	0.01										
4.0	-0.40	-0.07	0.36	0.43	0.28	0.13	0.05	0.02									
5.0	-0.18	-0.33	0.05	0.36	0.39	0.26	0.13	0.05	0.02								
5.53	0	-0.34	-0.13	0.25	0.40	0.32	0.19	0.09	0.03	0.01							
6.0	0.15	-0.28	-0.24	0.11	0.36	0.36	0.25	0.13	0.06	0.02							
7.0	0.30	0.00	-0.30	-0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02						
8.0	0.17	0.23	-0.11	-0.29	-0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03					
8.65	0	0.27	0.06	-0.24	-0.23	0.03	0.26	0.34	0.28	0.18	0.10	0.05	0.02				
9.0	-0.09	0.25	0.14	-0.18	-0.27	-0.06	0.20	0.33	0.31	0.21	0.12	0.06	0.03	0.01			
10.0	-0.25	0.04	0.25	0.06	-0.22	-0.23	-0.01	0.22	0.32	0.29	0.21	0.12	0.06	0.03	0.01		
12.0	0.05	-0.22	-0.08	0.20	0.18	-0.07	-0.24	-0.17	0.05	0.23	0.30	0.27	0.20	0.12	0.07	0.03	0.01

Observation from Bessel Function Table

- ❑ FM wave consists of the carrier {The first term in the equation}.
- ❑ FM wave consists of an infinite number of sidebands.
- ❑ The magnitude $V_c J_n$ of the carrier and sidebands depends on the Bessel Function.
- ❑ Some Bessel Function can be negative. Therefore, there is a 180 degree phase shift for that given sidebands.
- ❑ For Certain value of modulation index, the carrier component will disappear completely. These values are called eigen values.
- ❑ For FM Signal, Transmitted power always remains constant. It is not dependent on modulation index.

$$\Rightarrow P_t = \frac{V_c^2}{2R}$$

Bandwidth of FM wave

- ❑ Theoretically, the bandwidth of the FM wave is Infinite. But, practically, it is calculated based on how many sidebands have significant amplitude. The Simplest method to calculate the bandwidth is as under:

$$\Rightarrow BW = 2F_m \times \text{Number of Significant Sidebands}$$

- ❑ The second method is based on Carson's Rule.
- ❑ It States that the bandwidth of FM Wave is twice the sum of the deviation and highest modulating frequency.

$$\Rightarrow BW = 2 \Delta F + F_m$$

Parameters	DSB-FC	DSB-SC	SSB-SC	VSB
Bandwidth	❖ $2F_m$	❖ $2F_m$	❖ F_m	❖ $F_m + F_v$ ❖ 25% to 30% greater than SSB.

- ❑ The Bandwidth of FM is greater than AM, but FM has better noise immunity.

Example 1. A carrier wave of amplitude 10V and frequency 100 MHz is frequency modulated by a sinusoidal voltage. The modulating voltage has an amplitude of 5V and a frequency $f_m = 20$ kHz. The frequency deviation constant is 2 kHz/Volt. Draw the frequency spectrum of FM wave.

$$\begin{aligned}\leadsto V_c &= 10 \text{ V} \\ f_c &= 100 \text{ MHz} \\ V_m &= 5 \text{ V} \\ f_m &= 20 \text{ kHz} \\ K_F &= 2 \text{ kHz/V}\end{aligned}$$

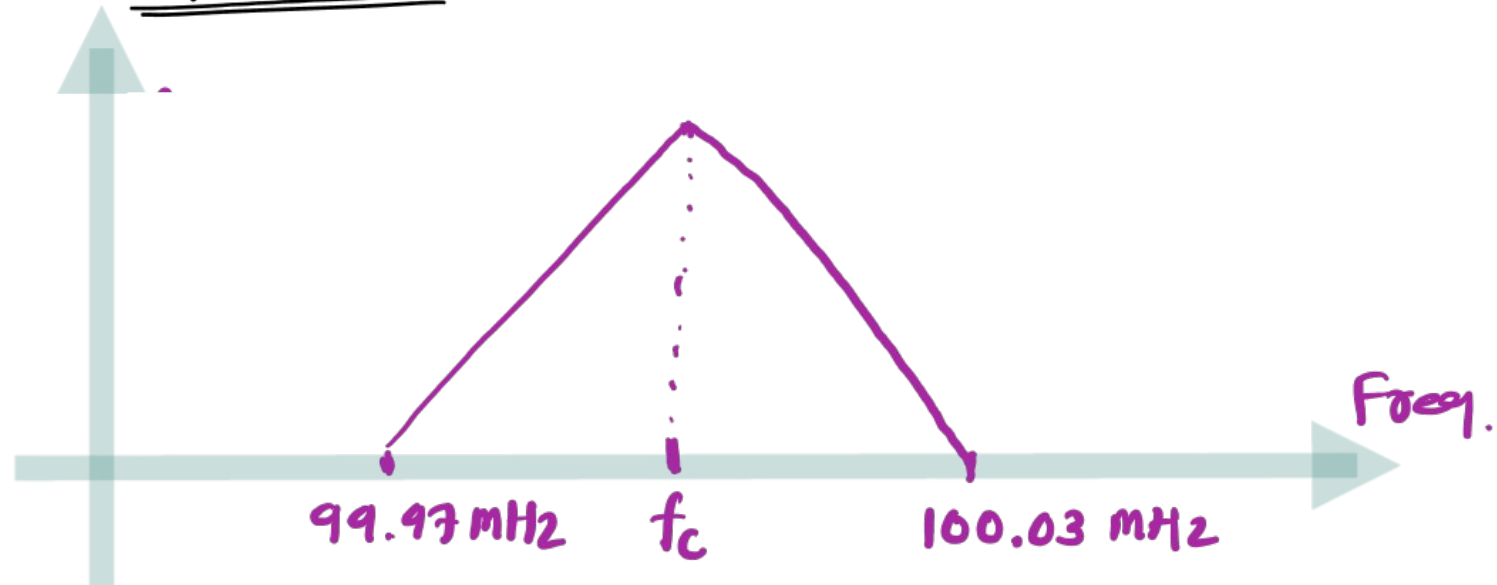
$$\begin{aligned}\rightarrow \Delta F &= K_F V_m \\ &= 2 \times 5 \\ &= 10 \text{ kHz}\end{aligned}$$

$\rightarrow$ Bessel Functions

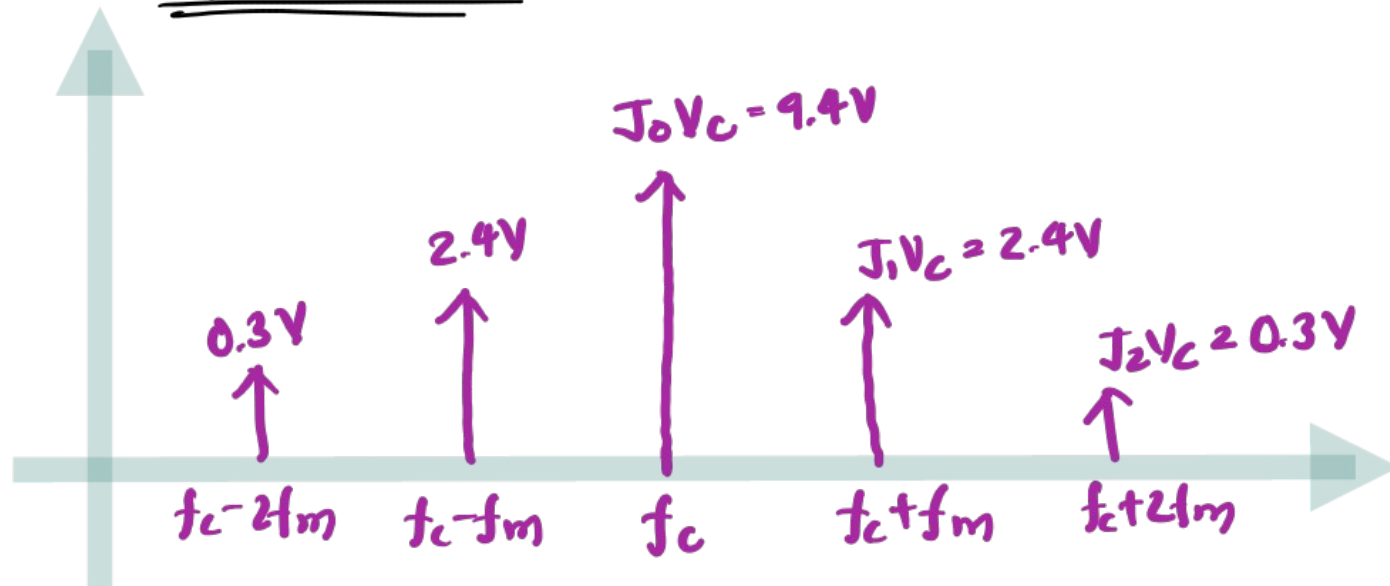
$$\begin{aligned}\rightarrow m_f &= \frac{\Delta F}{f_m} \\ &= \frac{10}{20} \\ &= 0.5\end{aligned}$$

$$\begin{aligned}\rightarrow BW &= 2 (f_m + \Delta F) \\ &= 2 (20 + 10) \\ &= 60 \text{ kHz}\end{aligned}$$

→ As per NBFM



→ As per Standard FM



Bessel Function Table with respect to Modulation Index

Modulation index	J_0	J_1	J_2	Sideband													
	Carrier	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0.00	1.00																
0.25	0.98	0.12															
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7.0	0.30	0.00	-0.30	-0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02						
8.0	0.17	0.23	-0.11	-0.29	-0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03					
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9.0	-0.09	0.25	0.14	-0.18	-0.27	-0.06	0.20	0.33	0.31	0.21	0.12	0.06	0.03	0.01			
10.0	-0.25	0.04	0.25	0.06	-0.22	-0.23	-0.01	0.22	0.32	0.29	0.21	0.12	0.06	0.03	0.01		
12.0	0.05	-0.22	-0.08	0.20	0.18	-0.07	-0.24	-0.17	0.05	0.23	0.30	0.27	0.20	0.12	0.07	0.03	0.01

Narrow Band Frequency Modulation (NBFM)

Outlines

- ❑ Basics of NBFM
- ❑ Derivation of the Equation of NBFM
- ❑ Essential Keynotes of NBFM
- ❑ NBFM Generation
- ❑ Resolution of Issue in NBFM Generation

Basics of NBFM

- NBFM is used to improve spectrum efficiency {Narrow Bandwidth}.
- NBFM is used in Voice Communication and Radio Settings.

{ Here, Modulation Index $\frac{\Delta F}{f_m}$ }

Derivation of the Equation of NBFM

Modulation Index for NBFM is less than 1.

Equation of Frequency Modulation {If the carrier signal is $V_c \cos \omega_c t$ and the Message Signal is $V_m \cos \omega_m t$ } is given by:

$$Y_{FM}(t) = V_c \cos \left(\omega_c t + \frac{k_f V_m}{\omega_m} \sin \omega_m t \right)$$

$$\Rightarrow (t) = V_c \cos \left(\omega_c t + m_f \sin \omega_m t \right)$$

$$Y_{FM}(t) = V_c \left[\cos \omega_c t \cos m_f \sin \omega_m t - \sin \omega_c t \sin m_f \sin \omega_m t \right]$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

- For Small values of Modulation Index, $\cos m_f \sin \omega_m t \approx 1$, $\sin m_f \sin \omega_m t \approx m_f \sin \omega_m t$

$$Y_{FM}(t) = V_c \cos \omega_c t - m_f V_c \sin \omega_c t \sin \omega_m t$$

$$\Rightarrow Y_{FM}(t) = V_c \cos \omega_c t + \frac{V_c}{2} \cos \left(\omega_c t - \omega_m t \right) - \frac{V_c}{2} \cos \left(\omega_c t + \omega_m t \right)$$

Carrier USB LSB

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)]$$

Essential Keynotes of NBFM

- Equation of NBFM {If the carrier signal is $V_c \cos \omega_c t$ and the Message Signal is $V_m \cos \omega_m t$ is given by:

$$Y_{FM}(t) = V_c \cos \left(\omega_c t + \frac{V_m}{V_c} \sin \omega_m t \right) = \frac{V_c}{2} \cos \omega_c t + \frac{V_c m_f}{2} \cos(\omega_c t - \omega_m t) - \frac{V_c m_f}{2} \cos(\omega_c t + \omega_m t)$$

Carrier USB LSB

$$\Rightarrow Y_{FM}(t) = V_c \cos \omega_c t - V_c m_f \sin \omega_c t \sin \omega_m t$$

$$\Rightarrow Y_{FM}(t) = V_c \cos \omega_c t - V_c m_f \sin \omega_c t \sin \omega_m t$$

$$\Rightarrow Y_{FM}(t) = V_c \cos \omega_c t - V_c m_f \sin \omega_c t \sin \omega_m t$$

Here, $g(t) = \int_0^t m(t) dt$

1. LSB is 180° out of phase with respect to USB.

2. Modulation Index is less than 1
 $m_f < 1$.

3. NBFM has only two side bands.

4. Amplitude of the sidebands is $\frac{V_c m_f}{2}$.

5. Frequency Deviation ΔF is limited in NBFM.

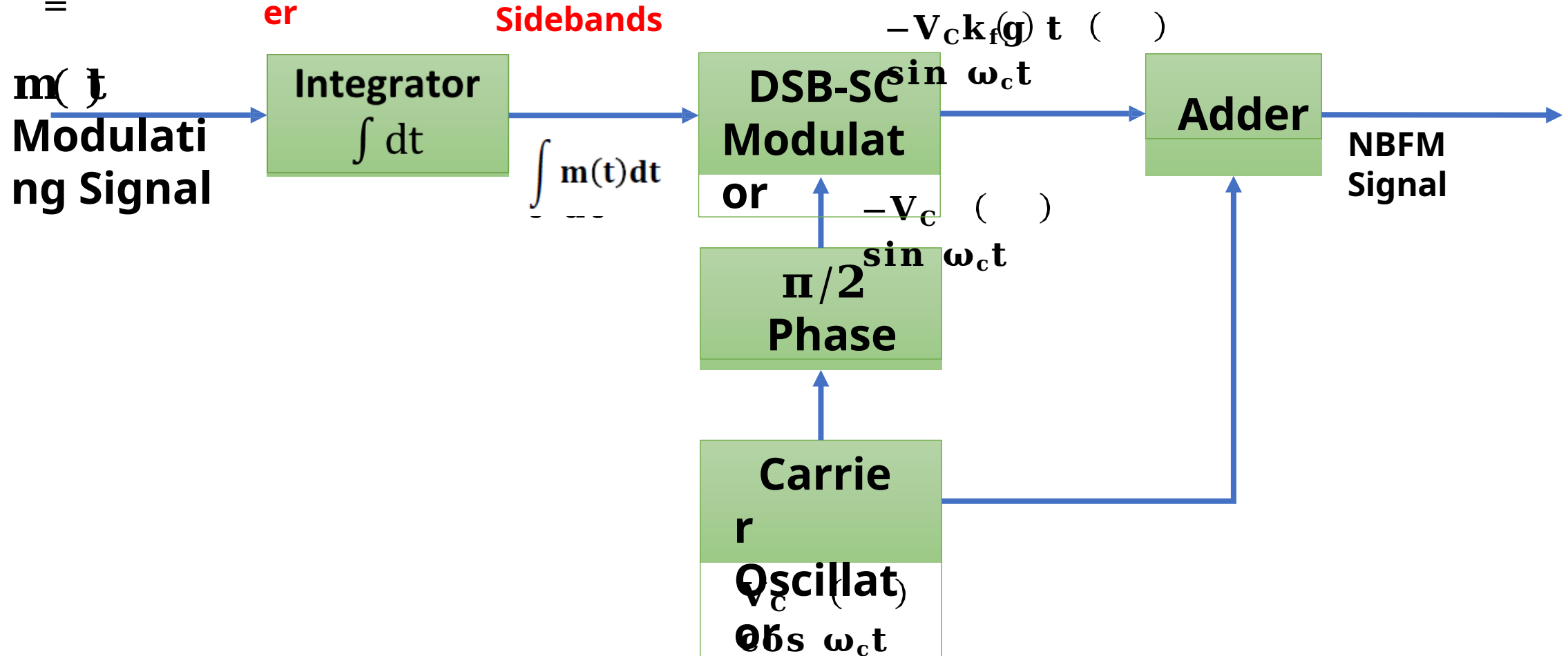
NBFM Generation

- Equation of NBFM {If the carrier signal is $V_c \cos \omega_c t$ and the Message signal is $m(t)$ is given by:

$$V_{FM} \cos \omega_c t = V_c \cos \left(\omega_c t + 2\pi k_f \int m(t) dt \right)$$

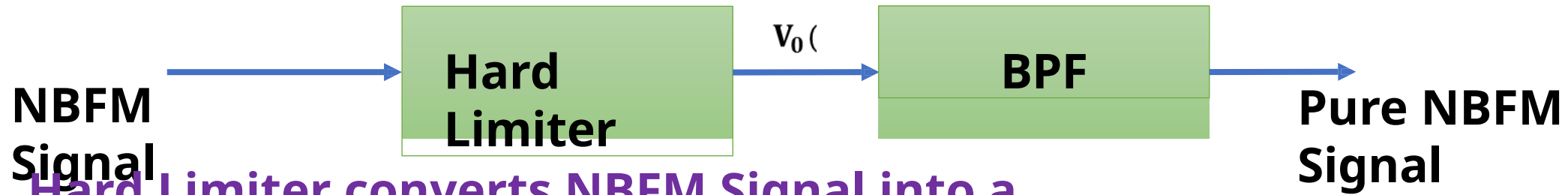
V_c Carrier
Sidebands

Here, $g(t) = \int_0^t m(t) dt$

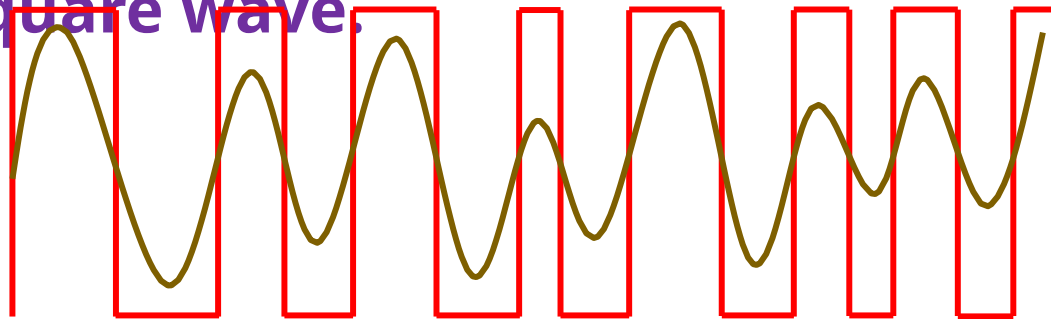


Resolution of Issue in NBFM Generation

- Amplitude variation can occur due to the adder. A Bandpass Limiter can resolve that issue.



- Hard Limiter converts NBFM Signal into a Square wave.



- The output of the Hard Limiter is

$$V_0(t) = \begin{cases} +1 & \cos \theta > 0 \\ -1 & \cos \theta < 0 \end{cases}$$

$$V_0(t) = \frac{1}{2} \cos \theta + \frac{1}{4} \cos 3\theta + \frac{1}{8} \cos 5\theta + \dots$$

- BPF will allow only $\cos \theta$ component.
- So, Pure NBFM Signal will be given by

$$\left\{ \text{Where, } \Rightarrow \theta = \omega_c t + 2\pi k_F \int m(t) dt \right\}$$

$$\text{Where, } \Rightarrow \theta = \omega_c t + 2\pi k_F \int m(t) dt$$

Example 2. The Signal $(\cos \omega_m t + 0.5 \cos \omega_m t) \sin \omega_c t$

A. FM only

B. AM only

☒ C. Both AM and FM

D. Neither AM nor FM

$\uparrow$
 f_c

is $\uparrow$
 $f_c \pm f_m$

$$\begin{aligned} \Rightarrow y_{AM} &= A_c (1 + m \sin \omega_m t) \sin \omega_c t \\ &= \underbrace{A_c \sin \omega_c t}_{f_c} + \underbrace{A_c m \sin \omega_m t \sin \omega_c t}_{f_c \pm f_m} \end{aligned}$$

$$\begin{aligned} \Rightarrow y_{NBFM} &= A_c (1 + K_f \int \omega_m t) \sin \omega_c t \\ &= \underbrace{A_c \sin \omega_c t}_{f_c} + \underbrace{K_f A_c \int \omega_m t \sin \omega_c t}_{f_c \pm f_m} \end{aligned}$$

Narrow Band Phase Modulation (NBPM)

Outlines

- ❑ Basics of NBPM
- ❑ Equation of NBPM
- ❑ NBPM Generation
- ❑ Resolution of Issue in NBPM Generation

Basics of NBPM

- NBPM is used to improve spectrum efficiency {Narrow Bandwidth} {It is similar to NBFM}.
- Modulation Index for NBPM is less than 1. {Here, Modulation Index m_p }
 $= k_p V_m$

Derivation of the Equation of NBPM

Equation of Phase Modulation {If the carrier signal is $V_c \cos \omega_c t$ and the Message Signal is $V_m \cos \omega_m t$ } is given by:

$\Rightarrow Y_{PM}(t) = V_c \cos(\omega_c t + (k_p)V_m \cos \omega_m t)$

$Y_{PM}(t)$

$$= V_c \cos(\omega_c t + m_p \cos \omega_m t)$$

$\Rightarrow$

$Y_{PM}(t)$

$$= V_c \{ \cos \omega_c t \cos m_p \cos \omega_m t - \sin \omega_c t \sin m_p \sin \omega_m t \}$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

- For Small values of Modulation Index, $\cos m_p \cos \omega_m t \approx 1$, $\sin m_p \sin \omega_m t \approx m_p \sin \omega_m t$

$$\Rightarrow Y_{PM}(t) = V_c \cos \omega_c t - m_p \sin \omega_c t \sin \omega_m t$$

$$= V_c \cos \omega_c t - V_c k_p V_m \sin \omega_c t \sin \omega_m t$$

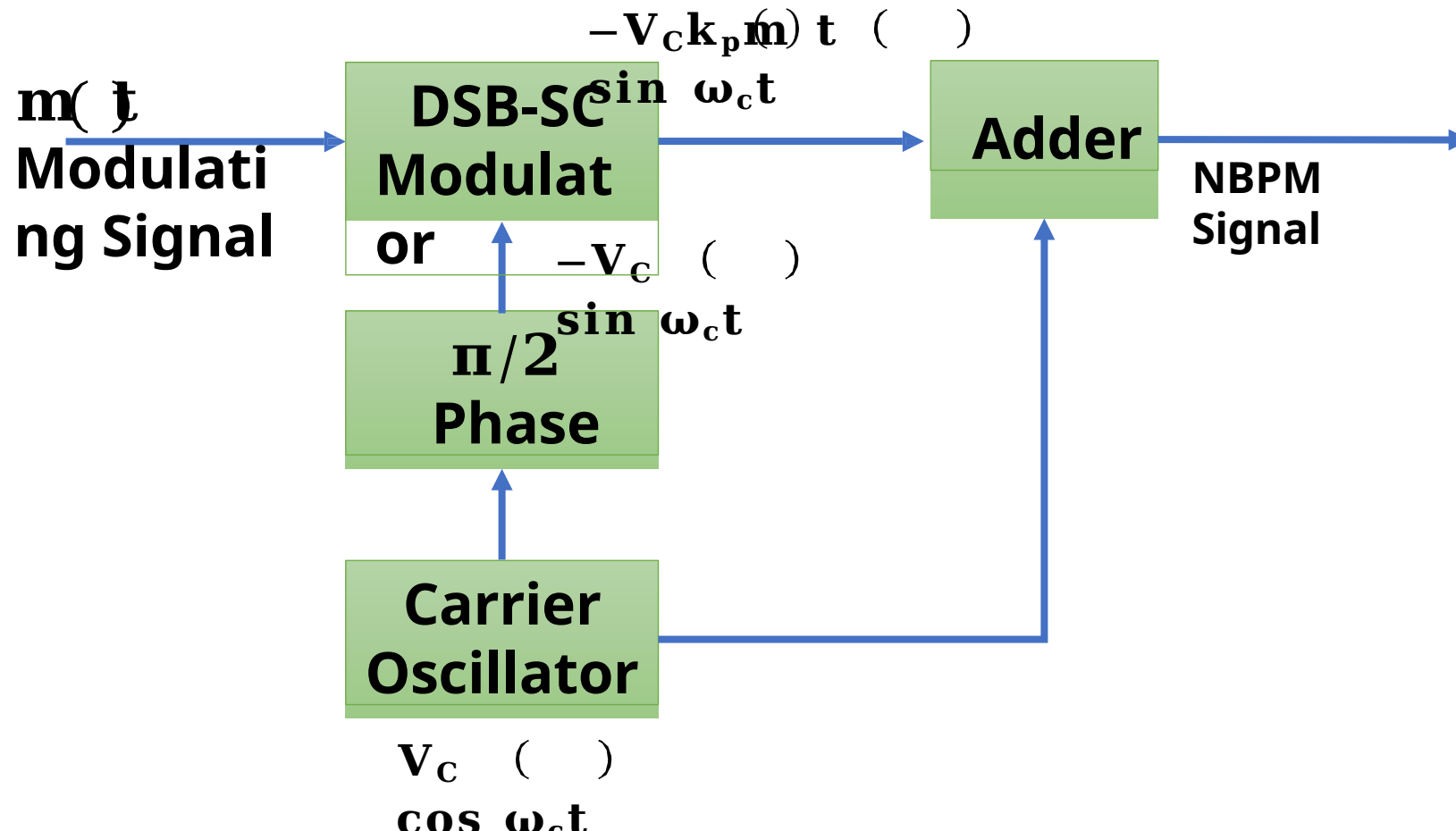
Carrier Sidebands

NBPM Generation

- Equation of NBPM {If the carrier is $V_c(\cos)\omega_c t$ and the Message Signal is $m(t)$ is given by

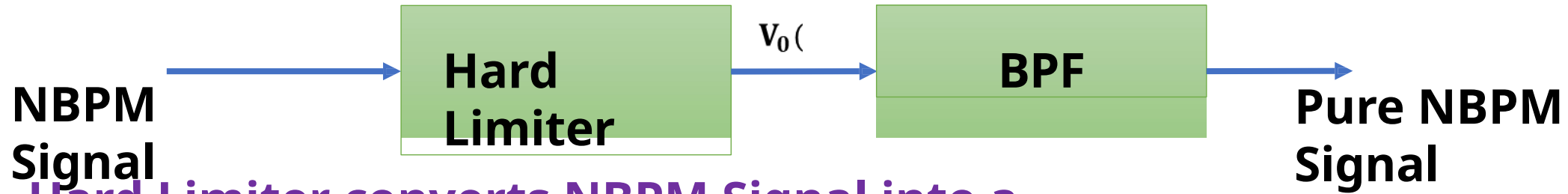
$$V_m = V_{PM} \cos \omega_c t + V_c k_p m(t) \sin \omega_c t$$

Carrier Sidebands

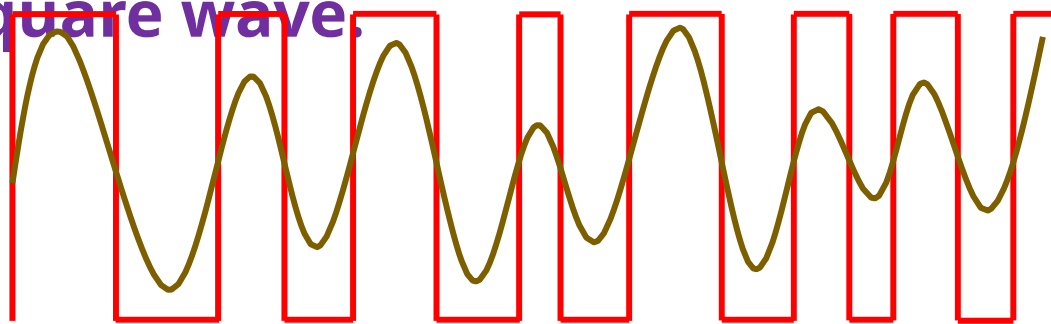


Resolution of Issue in NBPM Generation

- Amplitude variation can occur due to the adder. A Bandpass Limiter can resolve that issue.



- Hard Limiter converts NBPM Signal into a Square wave.



- The output of the Hard Limiter is

$$\Rightarrow V_0(t) = \begin{cases} +1 & \cos \theta > 0 \\ -1 & \cos \theta < 0 \end{cases}$$

$$\Rightarrow V_0(t) = \frac{4}{\pi} \left[\frac{1}{2} \cos \theta - \frac{1}{6} \cos 3\theta + \frac{1}{10} \cos 5\theta - \dots \right]$$

- BPF will allow only $\cos \theta$ component.
- So, Pure NBPM Signal will be given by:

$$\Rightarrow Y_{PM}(t) = \frac{4}{\pi} \cos \theta$$

$$\text{Where, } \theta = \omega_c t + k_{PM} m(t)$$

Wide Band Frequency Modulation (WBFM)

Outlines

- ❑ Basics of WBFM
- ❑ Bessel Function Table with respect to Modulation Index
- ❑ WBFM Generation
- ❑ Applications of WBFM

Basics of WBFM

- Modulation Index m_f for WBFM is Greater than 1

{ Here, Modulation Index m }
 ΔF

- The bandwidth of WBFM is several times the message Bandwidth.

- Equation of Frequency Modulation {If the carrier signal () and the Message

is $V_c \sin \omega_c t$

$$Y_{FM}(t) = V_c \sin(\omega_c t + m_f \sin(\omega_m t)) \text{ [Here, Modulation Index, } m_f = \frac{\Delta f}{f_m}]$$

Signal is $V_m \cos \omega_m t$ } is given by:

$$Y_{FM}(t) = V_c [J_0(m_f) \sin(\omega_c t) + J_1(m_f)(\sin((\omega_c + \omega_m)t) - \sin((\omega_c - \omega_m)t)) + J_2(m_f)(\sin((\omega_c + 2\omega_m)t) + \sin((\omega_c - 2\omega_m)t)) + J_3(m_f)(\sin((\omega_c + 3\omega_m)t) - \sin((\omega_c - 3\omega_m)t)) + J_4(m_f)(\sin((\omega_c + 4\omega_m)t) + \sin((\omega_c - 4\omega_m)t)) + \dots]$$

- In WBFM, as Modulation Index is greater than 1, Bessel Function J_n has more elements.

Bassel Function Table with respect to Modulation Index

Modulation index	Sideband																
	Carrier	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
0.00	1.00																
0.25	0.98	0.12															
0.5	0.94	0.24	0.03														
1.0	0.77	0.44	0.11	0.02													
1.5	0.51	0.56	0.23	0.06	0.01												
2.0	0.22	0.58	0.35	0.13	0.03												
2.41	0	0.52	0.43	0.20	0.06	0.02											
2.5	-0.05	0.50	0.45	0.22	0.07	0.02	0.01										
3.0	-0.26	0.34	0.49	0.31	0.13	0.04	0.01										
4.0	-0.40	-0.07	0.36	0.43	0.28	0.13	0.05	0.02									
5.0	-0.18	-0.33	0.05	0.36	0.39	0.26	0.13	0.05	0.02								
5.53	0	-0.34	-0.13	0.25	0.40	0.32	0.19	0.09	0.03	0.01							
6.0	0.15	-0.28	-0.24	0.11	0.36	0.36	0.25	0.13	0.06	0.02							
7.0	0.30	0.00	-0.30	-0.17	0.16	0.35	0.34	0.23	0.13	0.06	0.02						
8.0	0.17	0.23	-0.11	-0.29	-0.10	0.19	0.34	0.32	0.22	0.13	0.06	0.03					
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12.0	0.05	-0.22	-0.08	0.20	0.18	-0.07	-0.24	-0.17	0.05	0.23	0.30	0.27	0.20	0.12	0.07	0.03	0.01

WBFM Generation

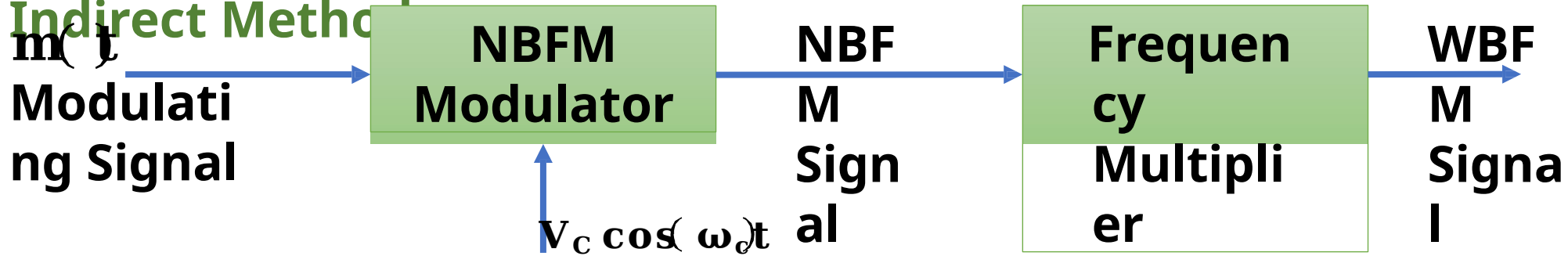
1. Direct Method



- VCO Produce Output whose frequency is proportional to input message signal $m(t)$. ()

$$\Rightarrow f_o = f_c + k_F m(t)$$

2. Indirect Method



- Frequency Multiplier produces signal whose frequency is n times of input frequency.

Applications of WBFM

- ❑ WBFM is widely used in FM Radio Broadcast {88 MHz to 108 MHz}.
 - FM Frequency Deviation is 75 KHz, for a 15 kHz Audio Band, there are 5 Bands.
- ❑ WBFM is used in a High-Fidelity Audio System.
- ❑ WBFM is used in Satellite Communication.
- ❑ WBFM is also used in Telemetry, which has a large data range.

FM Generation by Direct

Basics of FM Signal

- In Frequency Modulation, the Frequency of the Carrier Signal changes with respect to the Modulating Signal.

❖ Let the Carrier Signal is given by:
 by: $()$ $()$
 $\Rightarrow C(t) = V_C \cos \omega_c t$
 $= V_C \cos \theta(t)$

□ Then, the Frequency Modulated Signal is given
 $()$ $()$ $()$
 $() \Rightarrow Y_{FM}(t) = V_C \cos \omega(t) t$

❖ Here, $\omega(t)$ and $\theta(t)$ is a function of the Message Signal $m(t)$.

- Frequency Modulation is a part of the Angle Modulation, The Angle of Frequency

- The Frequency is given by:

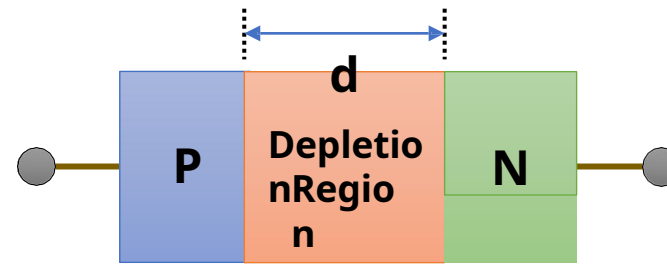
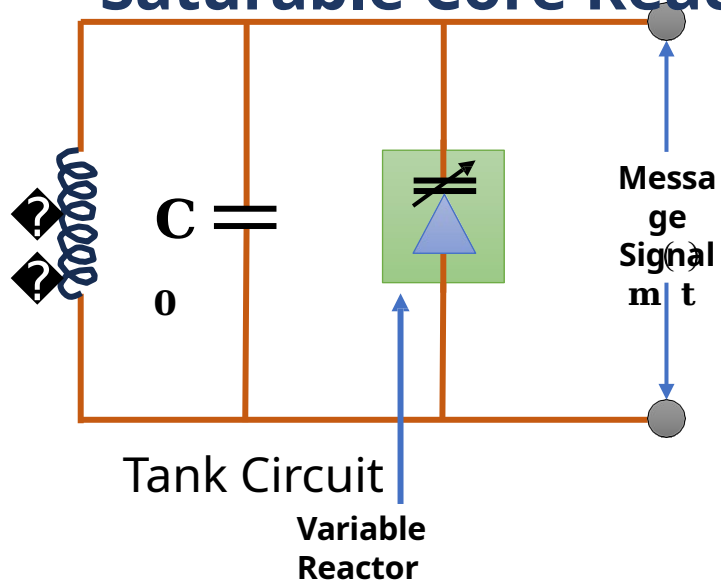
$$\Rightarrow \omega(t) = \omega_c + 2\pi k_f m(t)$$

- k_F is frequency Sensitivity, Unit is Hz/Volt.

Direct Method for FM Generation

- Using Voltage Controlled Oscillator (VCO), The frequency of Carrier signal can be controlled by an external voltage (Message Signal).

- Hartley or Colpitts Oscillator
- Schmitt Trigger Circuit
- Saturable Core Reactor



$$\epsilon_0 \epsilon_r \frac{C'}{d} = \frac{A}{d}$$

$$\Rightarrow C' = \frac{A}{d} \epsilon_0 \epsilon_r$$

- Effective Capacitance: ()

$$\Rightarrow C = C_0 + C' = C_0 + \frac{A}{d} \epsilon_0 \epsilon_r$$

Derivation of FM Signal using Direct Method for FM Generation

~> Oscillation Frequency

$$\Rightarrow \omega = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{L(C - k_m(t))}}$$

$$\Rightarrow \omega = \frac{1}{\sqrt{LC \left(1 - \frac{k_m(t)}{C_0}\right)}}$$

* Here, carrier freq. $\omega_c = \frac{1}{\sqrt{LC_0}}$

$$\Rightarrow \omega = \omega_c \left(1 - \frac{k_m(t)}{C_0}\right)^{-1/2}$$

* If, $\frac{k_m(t)}{C_0} \ll 1$, Higher orders of the Taylor's series will be neglected

$$\Rightarrow \omega = \omega_c \left(1 + \frac{k m(t)}{2\omega_c} \right)$$

$$\Rightarrow \omega = \omega_c + \frac{k\omega_c}{2\omega_c} m(t)$$

$$\Rightarrow \omega = \omega_c + k_f m(t)$$

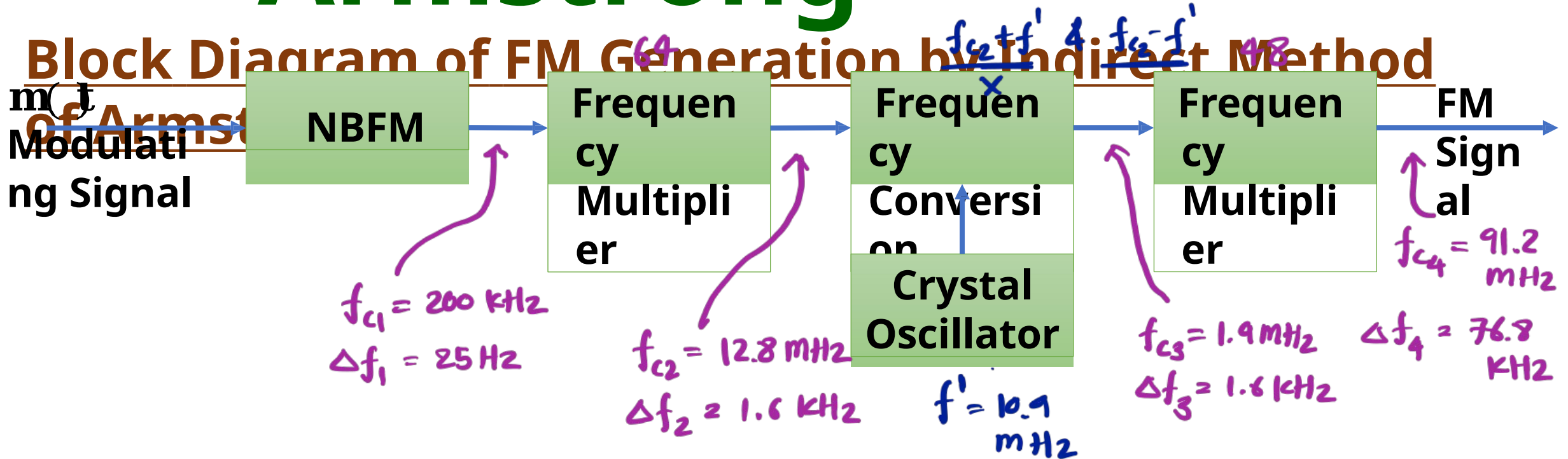
∴ Here, $k_f = \frac{k\omega_c}{2\omega_c} = \text{Frey. Sensitivity.}$

Advantages of Direct Method for FM Generation

- ❑ It generates sufficient frequency deviation.
- ❑ It requires little frequency multiplication.

Disadvantages of Direct Method for FM Generation

FM Generation by Indirect Method of Armstrong



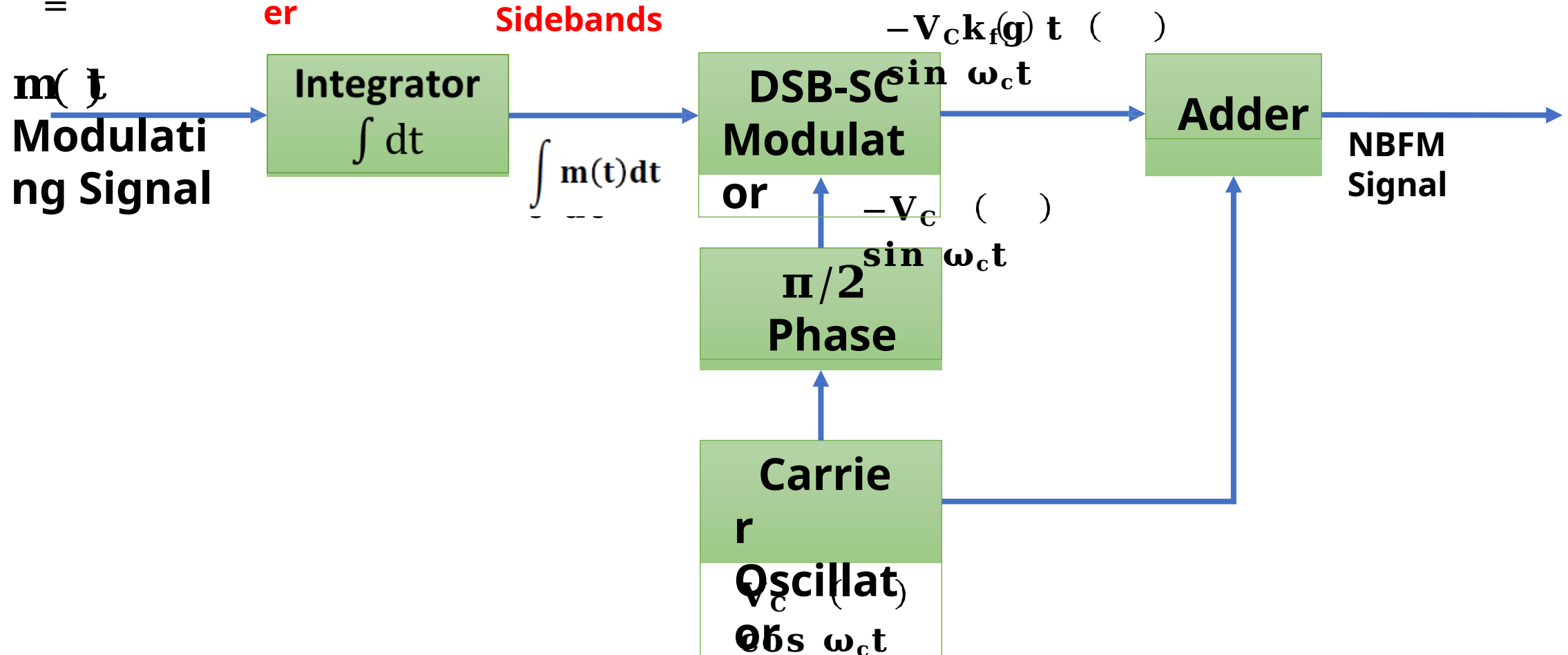
NBFM Generation

- Equation of NBFM {If the carrier signal is $V_c \cos \omega_c t$ and the Message signal is $m(t)$ is given by:

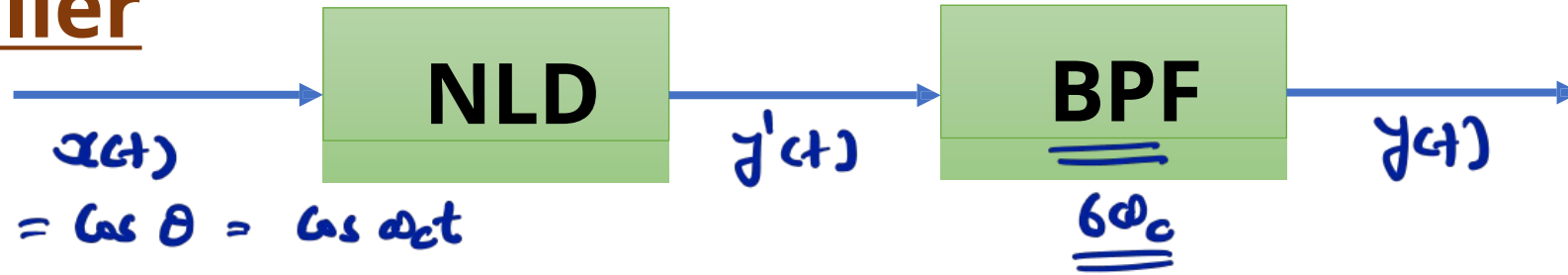
$$V_{FM} \cos \omega_c t = V_c \cos \left(\omega_c t + 2\pi k_f \int m(t) dt \right)$$

Carrier
Sidebands

Here, $g(t) = \int_0^t m(t) dt$



Frequency Multiplier



⇒ Output of NLD

$$\Rightarrow y'(t) = a_0 + a_1 x(t) + a_2 x^2(t) + a_3 x^3(t) + \dots$$

$$= a_0 + a_1 \cos \theta + a_2 \cos^2 \theta + a_3 \cos^3 \theta + \dots$$

$$- \cos^2 \theta = \frac{1}{2} (1 + \cos 2\theta)$$

$$- \cos^3 \theta = \frac{1}{4} (\cos \theta + \cos 3\theta)$$

$$\Rightarrow y'(t) = b_0 + b_1 \cos \theta + b_2 \cos 2\theta + b_3 \cos 3\theta + b_4 \cos 4\theta + \dots$$

$$= b_0 + b_1 \cos \omega_c t + b_2 \cos 2\omega_c t + b_3 \cos 3\omega_c t + b_4 \cos 4\omega_c t + \dots$$

⇒ Output of BPF $\Rightarrow y(t) = b_6 \cos 6\omega_c t$

FM

Outlines

- ❑ FM Demodulation using Envelope Detector
- ❑ Waveforms of FM Demodulation using Envelope Detector

Demodulation

FM Demodulation using Envelope Detector



⇒ FM Signal.

$$\Rightarrow y_{FM}(t) = V_c \cos(\omega_c t + 2\pi K_f \int m(t) dt)$$

⇒ Output of differentiator.

$$\Rightarrow y'(t) = \frac{d}{dt} y_{FM}(t) = V_c \frac{d}{dt} \cos(\omega_c t + 2\pi K_f \int m(t) dt)$$

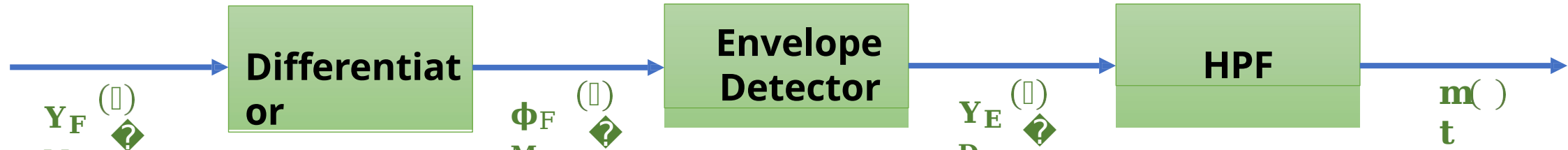
$$\Rightarrow y'(t) = -V_c \underbrace{(\omega_c + 2\pi K_f m(t))}_{\text{Envelope}} \sin(\omega_c t + 2\pi K_f \int m(t) dt)$$

⇒ Cond.ⁿ for Envelope detector. $[\omega_c > 2\pi K_f m(t)]$

⇒ Output of Envelope detector.

$$\Rightarrow y''(t) = \underbrace{\omega_c}_{\text{DC}} + \underbrace{2\pi K_f m(t)}_{\text{Message Signal.}}$$

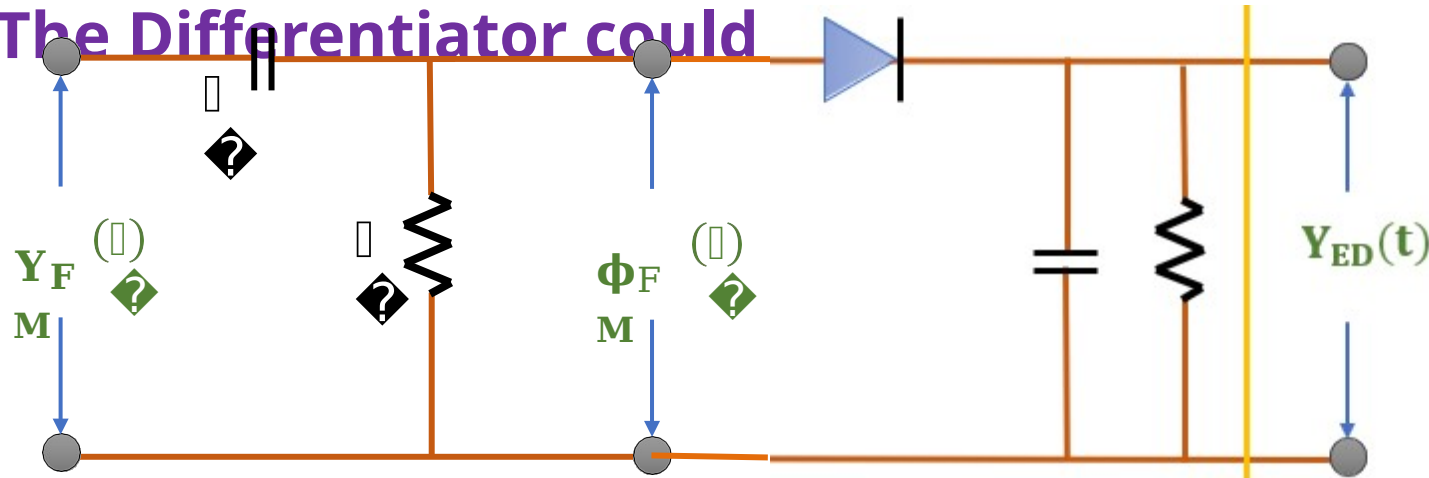
Circuit Diagram



- The Transfer Function of Differentiator H_f is directly Proportional to Frequency F .

$$\Rightarrow H_f \propto f$$

- The Differentiator could be implemented using an Op Amp and a High Pass Filter HPF.



- The Transfer Function is given by:

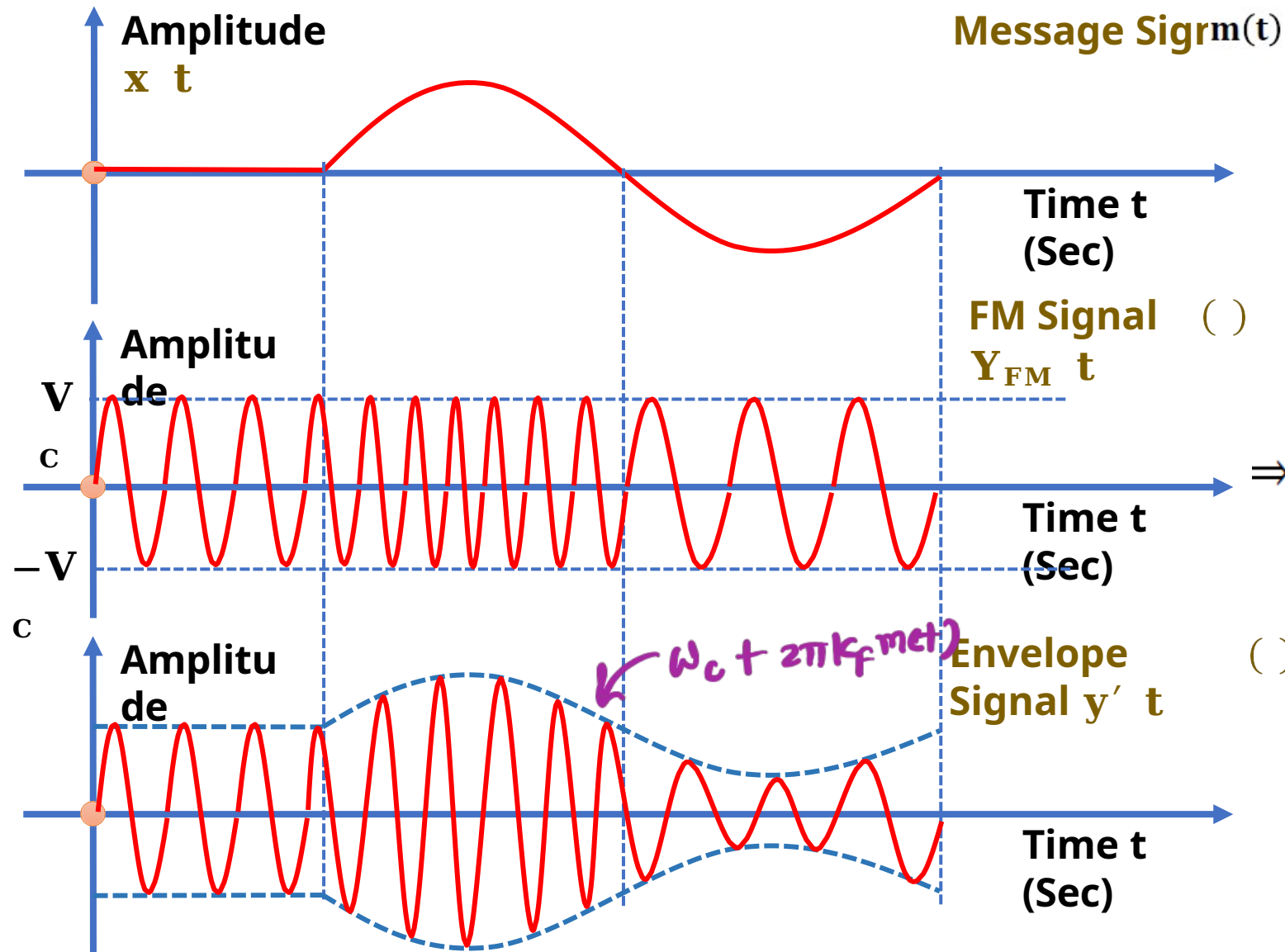
$$\Rightarrow H_f = \frac{\Phi_F(f)}{Y_F(f)} = \frac{R}{j\omega CR + 1}$$

- If $\omega RC \ll 1$, then

$$\Rightarrow H_f = \frac{R}{j\omega CR}$$

▪ Here, The differentiator Slope $\frac{dY_{FM}}{dt} =$

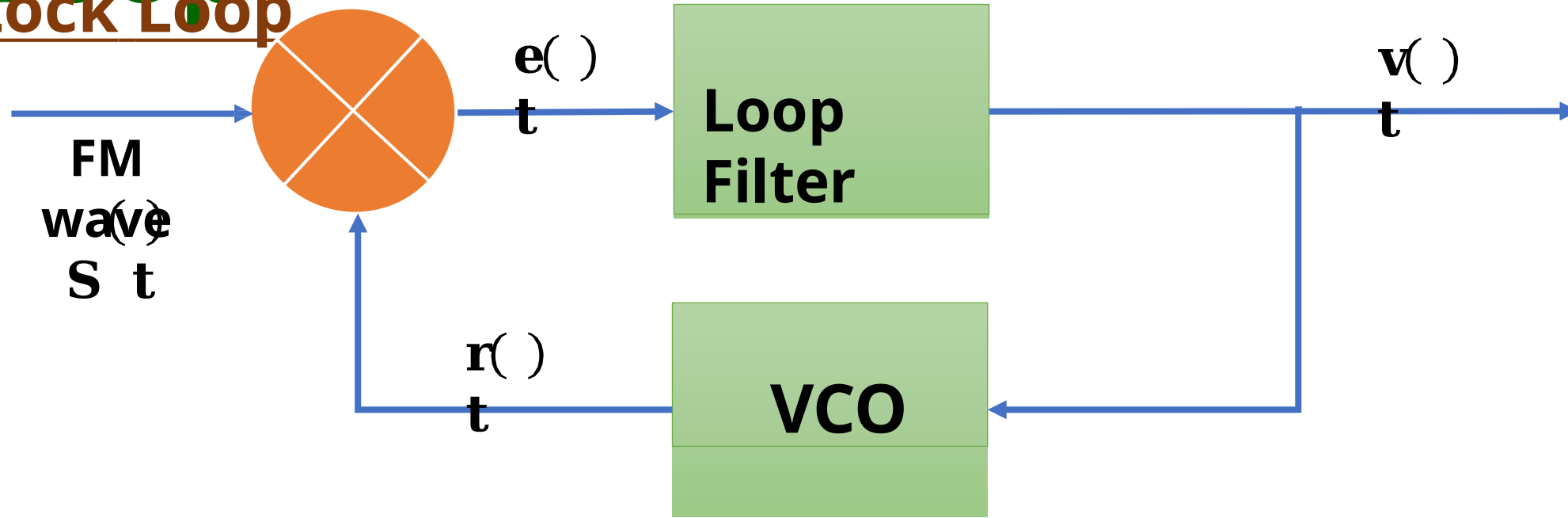
Waveforms of FM Demodulation using Envelope Detector



$$\Rightarrow Y_{FM}(t) = V_c \cos \left(\omega_c t + 2\pi k_F \int_0^t m(t) dt \right)$$

FM Demodulation by Phase Lock

Block Diagram of Phase Lock Loop



□ The Phase Lock Loop has three essential elements with Negative feedback:

- **Multiplier:** It is used as a Phase Detector.
- **VCO:** It works as a Local Oscillator.
- **Loop Filter:** It acts as a Low Pass Filter.

⇒ FM signal $s(t)$

$$\Rightarrow s(t) = V_c \cos(\omega_c t + \phi_1(t))$$

$$\text{where, } \phi_1(t) = 2\pi K_F \int_0^t m(t) dt$$

⇒ Output of VCO $r(t)$

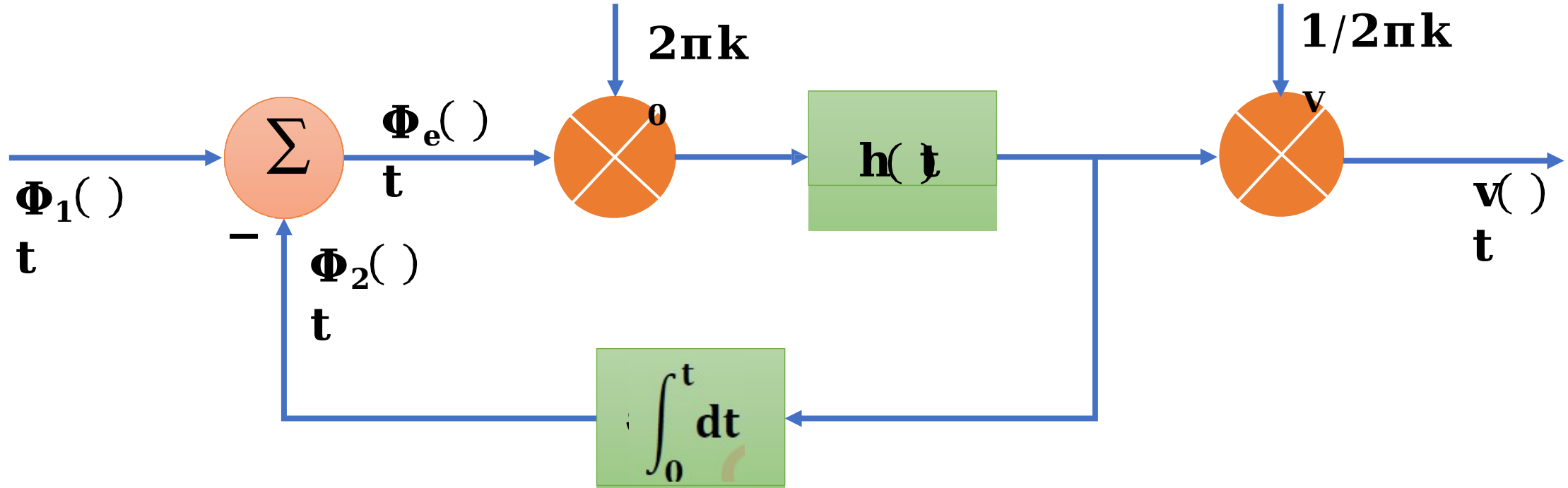
$$\Rightarrow r(t) = V_c \cos(\omega_c t + \phi_2(t))$$

$$\text{where, } \phi_2(t) = 2\pi K_V \int_0^t v(t) dt$$

⇒ Output of Multiplier

$$\Rightarrow e(t) = s(t)r(t)$$

Linear Model of Phase Lock Loop



$\Rightarrow$ Phase lock cond.ⁿ $\Rightarrow \Phi_e(t) = 0$

$$\Rightarrow \Phi_e(t) = \Phi_1(t) - \Phi_2(t) = 0$$

$$\Rightarrow \Phi_1(t) = \Phi_2(t)$$

$$\Rightarrow 2\pi k_F \int_0^t m(t) dt = 2\pi k_V \int_0^t v(t) dt$$

⇒ Differentiate above eq.ⁿ with respect to time

$$\Rightarrow m(t) = \frac{k_v}{k_f} v(t)$$

$$\Rightarrow m(t) \propto v(t)$$

→ So, Output is demodulated as message signal.